Seunghyun Lee

Curriculum Vitae

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Research Interests

Algebraic Number Theory, Geometric Measure Theory, Harmonic Analysis

Education

MSci in Mathematics, Imperial College London — Sep 2023-Jun 2027

- Grade: First Class Honours
- Dissertation: *On the disproof of the Erdös distance problem*, supervised by Prof. Toby Gee
- Modules: Representation Theory, Functional & Fourier Analysis, Algebraic Topology (Year 3), Elliptic Curves, Algebraic Geometry, Complex Manifolds, Homological Algebra (Year 4)

Publications & Preprints

1. *Classification of λ-translators in the Lorentz-Minkowski space* with Marie-Amélie Lawn, Miguel Ortega, in preparation.

Research Experience

- Final Year Dissertation, Imperial College London — Oct 2026-Present
  On the disproof of the Erdös distance problem
- Summer Research Programme (UROP), Imperial College London — Jul 2025-Sep 2025
  λ-translators in the Lorentz-Minkowski space
- 2nd Year Research Project, Imperial College London — May 2025-Jun 2025
  The endpoint case of the multilinear Kakeya estimate
- 1st Year Research Project, Imperial College London — May 2024-Jun 2024
  Finding eigenvalues and eigenfunctions of compact operators

Talks

4. DRP Symposium, Imperial College London — Jan 2026
3. 3-minute UROP Thesis Competition, Imperial College London — Oct 2025
2. 2nd Year Research Symposium, Imperial College London — Jun 2025
1. 1st Year Research Symposium, Imperial College London — Jun 2024

Honours

- Dean's List (Years 1 & 3), Imperial College London — Oct 2024, Oct 2026
- Best Presentation Award, Imperial College London — Oct 2025
- UROP Prize, Imperial College London — Oct 2025
- UROP Departmental Bursary, Imperial College London — May 2025

Teaching

Imperial College London

- Research Mentor, Directed Reading Programme (DRP) — Autumn 2026-Present
  Fourier analysis on finite groups and applications to additive combinatorics
- 1st Year Peer Tutor — Autumn 2025-Present
  Analysis 1, Linear Algebra and Groups, Calculus and Applications, Probability and Statistics
- Research Mentor, Directed Reading Programme (DRP) — Autumn 2025
  Hausdorff dimension and Kakeya sets

Conferences & Workshops Attended

- Undergraduate Summer School, London Mathematical Society — Jul 2025
  Knot theory, additive number theory, topological solitons, sub-Riemannian geometry

Skills

Programming: Python, LaTeX (Proficient); C++, R, MATLAB (Familiar)

Languages: Korean (Native), English (Fluent).